SIA SOFTWARE DEVELOPMENT · GROUP 4 DELIVERABLE

STORA: Draw It, Then Prove It

Comprehensive Speaker Notes, Technical Defense Guide, Master Glossary of Technical Terms, and Slide-by-Slide Explanations for SIA Technical Panel Defense.

PROJECT	TOPICS COVERED	DECK SIZE	LIVE ARTEFACTS
STORA Retail OS	13, 14, 15, 16	19 Widescreen Slides	6 ADRs, UAT, A11Y Audit

📖 Master Glossary: Technical Terms, Full Words & Their Purpose

Quick-reference guide explaining all technical abbreviations and concepts used throughout the presentation, defense deck, and STORA codebase.

UML **Unified Modeling Language**

Purpose: A standardized visual language in software engineering to create blueprints of architecture, user interactions, and databases so programmers and business stakeholders share identical understanding.

C4 Model **Context, Containers, Components, Code**

Purpose: An architecture diagramming framework (by Simon Brown) structured like Google Maps. It zooms from high-level ecosystem (Context) down to individual code classes without overwhelming non-technical evaluators.

ADR **Architecture Decision Record**

Purpose: A short, immutable document capturing a major technical choice, its problem context, and trade-offs. Preserves institutional memory so teams never forget why a technology was picked months earlier.

UAT **User Acceptance Testing**

Purpose: The final pre-release testing phase where real end-users (sari-sari store owners and shoppers) perform real-world tasks to verify whether the system is genuinely intuitive and solves practical problems.

RA 10173 **Republic Act No. 10173 (Data Privacy Act of 2012)**

Purpose: The Philippine legal standard governing data protection. Mandates informed consent, data minimization, purpose specification, and user rights (right to be forgotten/deleted) during software testing and production.

a11y **Accessibility (11 letters between 'a' and 'y')**

Purpose: The engineering discipline of building apps usable by everyone, including people with vision impairments, motor limitations, color blindness, or outdoor conditions like glare under tropical sunlight.

WCAG **Web Content Accessibility Guidelines**

Purpose: The international benchmark (W3C) establishing testable standards for accessibility, such as a 4.5:1 minimum text contrast ratio and 44×44dp minimum touch target dimensions.

ERD **Entity-Relationship Diagram**

Purpose: A visual database schema model defining entities (tables), fields, and relationships (one-to-many, foreign keys) ensuring structural integrity before writing SQL migrations.

ACID **Atomicity, Consistency, Isolation, Durability**

Purpose: Four fundamental relational database transaction guarantees that prevent financial ledger corruption. In STORA, ensures store credit (*utang*) transactions either succeed completely or roll back safely.

POS **Point of Sale**

Purpose: The retail transaction interface where merchants scan items, calculate change (*sukli*), record payments, and deduct inventory in real time.

REST API **Representational State Transfer App Programming Interface**

Purpose: A standardized, stateless HTTP architectural style allowing mobile clients (Flutter) and backend servers (Django) to exchange data cleanly using JSON payloads and HTTP status codes.

JWT **JSON Web Token**

Purpose: A cryptographically signed token enabling stateless user authentication between the mobile phone and backend server without database lookups on every single HTTP request.

FCM **Firebase Cloud Messaging**

Purpose: Google's cloud notification service that wakes up mobile devices to show push alerts (e.g., when a customer places an order or a merchant confirms preparation).

SIA **Systems Integration and Architecture**

Purpose: The overarching software engineering discipline that ensures separate subsystems (databases, mobile frontends, cloud servers, and human workflows) integrate harmoniously into a unified product.

SLIDE 01 TITLE & OVERVIEW

Foundation

Draw it, then prove it

🎯 SLIDE PURPOSE & CORE MEANING

Establishes the central thesis: **Running code is not enough**. A complete software system must be clearly diagrammed so others can understand it, technical choices must be justified with written evidence, and usability must be proven with real sari-sari store owners and shoppers under Philippine law.

🎙️ PRESENTER SCRIPT (WHAT TO SAY)

> **VERBATIM SPOKEN GUIDE:**
> *"Good morning / afternoon members of the panel. Today, Group 4 presents 'Draw it, then prove it' for STORA—our smart inventory and Point of Sale (POS) platform. Anyone can write code that runs on their own machine, but Systems Integration and Architecture (SIA) maturity requires proving three things: First, can we explain our architecture in seconds using Unified Modeling Language (UML) and the C4 Model? Second, can we justify why we built it this way through Architecture Decision Records (ADRs)? And third, can real community sari-sari store owners actually use it under Philippine Data Privacy law (RA 10173)? Over the next 18 slides, we will prove all three."*

💡 KEY TAKEAWAY FOR THE PANEL

The four core pillars of this unit are: **C4 & UML Diagrams (Topic 13)**, **Architecture Decision Records (Topic 14)**, **User Acceptance Testing & RA 10173 Consent (Topic 15)**, and **Accessibility & Go-Live (Topic 16)**.

SLIDE 02 WHAT GROUP 4 OWNS

Team Responsibility

Four topics: make the system legible, then validated

🎯 SLIDE PURPOSE & CORE MEANING

Defines where Group 4 sits in the 5-group software engineering pipeline. Groups 1–3 built the backend, database, and client apps. Group 4's job is not rewriting their code, but acting as the critical quality and documentation bridge before Group 5 deploys it.

🎙️ PRESENTER SCRIPT (WHAT TO SAY)

> **VERBATIM SPOKEN GUIDE:**
> *"To understand our role, look at the pipeline: Groups 1 through 3 constructed the engine—the Application Programming Interfaces (APIs), relational models, and Flutter screens. Group 4 owns system legibility and human validation. We produce two architecture deliverables (C4/UML diagrams and Architecture Decision Records) and two human-centered deliverables (User Acceptance Testing with signed legal consent and Accessibility compliance). Group 5 cannot deploy to production until our criteria are 100% satisfied."*

> ⚠️ **POTENTIAL PANEL QUESTION & BULLETPROOF ANSWER:**
> *"Why does Group 4 spend time on documentation instead of building more features?"*
> **ANSWER:**
> "Because undocumented software creates massive technical debt. If a team member leaves, unrecorded decisions lead to expensive rewrites. Group 4 ensures STORA is maintainable, auditable, and proven before public release."

SLIDE 03 STRATEGIC VALUE

Philosophy

A system nobody can explain is one nobody can trust

🎯 SLIDE PURPOSE & CORE MEANING

Breaks down the three essential verification verbs: **SHOW IT** (pictures beat 1,000 words), **JUSTIFY IT** ("Why did you...?" has a documented answer), and **PROVE IT** (real users with disabilities and low tech-literacy, not internal team members).

🎙️ PRESENTER SCRIPT (WHAT TO SAY)

> **VERBATIM SPOKEN GUIDE:**
> *"A working system nobody can explain is a black box that nobody can trust. When an evaluator or investor asks, 'How does your checkout handle network failure?', talking for ten minutes in prose confuses the room. A single C4 container diagram answers it in 30 seconds. When they ask, 'Why didn't you use MongoDB?', pulling up our ADR-0002 proves our choice was intentional. And when they ask if it is user-friendly, our opinion doesn't matter—our recorded User Acceptance Testing (UAT) data does."*

SLIDE 04 TOPIC 13 SECTION HEADER

Architecture

Topic 13: C4 & UML Diagrams

🎯 SLIDE PURPOSE & CORE MEANING

Introduces the diagramming discipline. The most important concept is **zoom level**: giving the right amount of detail to the right audience without drowning them in code trivialities.

🎙️ PRESENTER SCRIPT (WHAT TO SAY)

> **VERBATIM SPOKEN GUIDE:**
> *"Now we move to Topic 13: C4 and UML diagrams. The golden rule of software modeling is: a diagram is only useful at the right zoom level. You would never hand a circuit board diagram to someone trying to navigate a highway, and you wouldn't give a high-level marketing graphic to a database administrator. We use the C4 model to map STORA cleanly across four distinct zoom levels."*

SLIDE 05 TOPIC 13: C4 MODEL

Deep-Dive

Four zoom levels — show only what the audience needs

🎯 SLIDE PURPOSE & CORE MEANING

Explains the four zoom levels of Simon Brown's C4 model (Context, Container, Component, Code) and how STORA maps to each:

- **1. System Context:** STORA as a single box surrounded by Store Owners, Shoppers, GCash QR, and Firebase Cloud Messaging (FCM). For non-technical stakeholders.
- **2. Container:** High-level deployable units: `stora_owner` (Flutter), `stora_customer` (Flutter), Django REST API, and PostgreSQL on Render.
- **3. Component:** Inside the backend: Authentication handler, Inventory serializer, POS Checkout service, and FCM dispatchers.
- **4. Code:** Individual class hierarchies and methods (auto-generated by IDE, omitted from manual hand-drawing).

🎙️ PRESENTER SCRIPT (WHAT TO SAY)

> **VERBATIM SPOKEN GUIDE:**
> *"Think of C4 like Google Maps: Level 1 (Context) is the continent view—who uses STORA and what third-party systems like GCash and Firebase touch it? Level 2 (Container) is the city view—what are our deployable containers? We have two Flutter mobile clients communicating over HTTPS with a Django REST backend and a managed PostgreSQL database. For our defense, Levels 1 and 2 are all the panel needs to understand our entire operational landscape."*

> 💡 **STORA ARCHITECTURE REFERENCE:**
> See Section 6 of `ARCHITECTURE.md` for the exact Mermaid container diagram linking the dual Flutter clients with the Django REST service.

SLIDE 06 TOPIC 13: UML SELECTION

UML Modeling

Three UML diagrams earn their keep — pick by the question

🎯 SLIDE PURPOSE & CORE MEANING

Rejects the antipattern of drawing all 14 Unified Modeling Language (UML) diagrams just to fill pages. Identifies the three diagrams that actually provide value:

- **Use-Case Diagram ("Who can do what?"):** Maps store owner actions (barcode scanning, stock adjustments, utang tracking) vs customer actions (order placement, store radar).
- **Sequence Diagram ("What talks to what, in what order?"):** Essential for tracing tricky financial flows like Fast Checkout: Client ➔ Local Cache ➔ Django ➔ Postgres Stock Lock ➔ FCM.
- **Class / Entity-Relationship Diagram (ERD) ("How is data structured?"):** Maps relational tables: User, StoreLocation, Product, Category, Order, and UtangLedger.

🎙️ PRESENTER SCRIPT (WHAT TO SAY)

VERBATIM SPOKEN GUIDE:

"*Instead of drawing every UML diagram, we focus on three that earn their keep: Use-Case sets our user scope; Sequence traces our multi-step checkout and stock deduction; and our Entity-Relationship Diagram (ERD) reflects the real PostgreSQL schema created by Group 3. Most importantly: our diagrams are living documents that reflect our actual working code.*"

⚠️ **CRUCIAL RULE:**

"A diagram that does not match the code is worse than no diagram at all. It misleads future maintainers. Our ERD matches our Django migrations 1-to-1."

SLIDE 07 TOPIC 14 SECTION HEADER

ADR Foundation

Topic 14: Architecture Decision Records (ADRs)

🎯 SLIDE PURPOSE & CORE MEANING

Introduces Architecture Decision Records. The key insight: **Diagrams show what you built; ADRs capture why you built it.**

🎙️ PRESENTER SCRIPT (WHAT TO SAY)

> **VERBATIM SPOKEN GUIDE:**
> *"Moving to Topic 14: Architecture Decision Records, or ADRs. When you present an app, the panel will inevitably ask: 'Why this framework? Why this database?' Without ADRs, teams invent answers on the fly. An ADR is an immutable record that freezes our architectural reasoning at the moment we made the choice."*

SLIDE 08 TOPIC 14: ADR FORMAT

ADR Standard

One short page per decision — four headings, that is all

🎯 SLIDE PURPOSE & CORE MEANING

Deconstructs the standard Michael Nygard Architecture Decision Record format into four mandatory sections:

- **1. Status:** Proposed, Accepted, or Superseded.
- **2. Context:** The technical problem, environmental constraints (e.g. Philippine 3G/4G connectivity, offline store realities).
- **3. Decision:** The explicit, unambiguous choice made.
- **4. Consequences:** Both benefits gained and trade-offs/costs accepted.

🎙️ PRESENTER SCRIPT (WHAT TO SAY)

> **VERBATIM SPOKEN GUIDE:**
> *"An ADR is not a 50-page dissertation. It is exactly one page with four headings: Status, Context, Decision, and Consequences. Look at ADR-0001: We chose Django REST Framework over GraphQL. Why? Because sari-sari stores operate on throttled prepaid cellular data. REST enables standard HTTP caching headers and is faster to inspect with standard tooling. If our requirements change later, we don't delete this ADR—we write a new one that supersedes it."*

> 💡 **STORA IMPLEMENTATION:**
> All ADRs are stored as numbered markdown files in `docs/adr/` (e.g. `0001-use-django-rest-api.md`) versioned directly inside Git.

SLIDE 09 TOPIC 14: ADRS IN PRACTICE

Practical Rules

Record the decisions that were hard to make or hard to reverse

🎯 SLIDE PURPOSE & CORE MEANING

Provides the acid test for what warrants an ADR vs what is trivial clutter. Lists STORA's 6 production ADRs:

- **ADR-0001:** REST over GraphQL (predictable caching & network efficiency).
- **ADR-0002:** PostgreSQL over NoSQL (strict ACID compliance for community debt / utang ledgers).
- **ADR-0003:** Dual Flutter Apps (independent Owner and Customer APKs for fast install times and clean role isolation).
- **ADR-0004:** Drift/SQLite Local Caching (guarantees offline POS checkout resilience during power outages).
- **ADR-0005:** RA 10173 Compliance (multi-tenant role isolation and permanent account deletion cascading).
- **ADR-0006:** Hybrid JWT + Firebase Auth (decoupled session tokens with cloud push notifications).

🎙️ PRESENTER SCRIPT (WHAT TO SAY)

> **VERBATIM SPOKEN GUIDE:**
> *"What deserves an ADR? Anything that would take weeks to change if we were wrong. For instance, choosing PostgreSQL over MongoDB is an ADR because sari-sari store credit—utang—demands strict ACID transactional consistency; you cannot have phantom balances or partial debits. Conversely, variable names or button colors do not get ADRs. We have 6 official ADRs documented in docs/adr/ covering our entire architectural core."*

SLIDE 10 TOPIC 15 SECTION HEADER

Human Validation

Topic 15: User Acceptance Testing (UAT) & Consent

🎯 SLIDE PURPOSE & CORE MEANING

Transition from automated machine testing to human usability verification. Automated unit tests verify that the code executes correctly; UAT verifies that the software is intuitive and valuable to real human beings.

🎙️ PRESENTER SCRIPT (WHAT TO SAY)

> **VERBATIM SPOKEN GUIDE:**
> *"Moving to Topic 15: User Acceptance Testing, or UAT. Automated unit tests answer: 'Did we build the system right?' But UAT answers a far harder question: 'Did we build the right system for the user?' Even if all 37 Flutter tests pass, the app is a failure if an elderly sari-sari owner cannot figure out how to scan a barcode or ring up change."*

SLIDE 11 TOPIC 15: UAT PROTOCOL

Testing Methodology

Give real users a real task, then be quiet and watch

🎯 SLIDE PURPOSE & CORE MEANING

Details the **Silent Observation Protocol** and pre-agreed acceptance criteria:

- **Target Cohort:** 5 real sari-sari store owners and 5 neighborhood shoppers (never test with the developers who wrote the code).
- **Realistic Tasks:** "Scan 2 products and ring up a cash payment with ₱500 bill" or "Find a neighborhood store and order pan de sal."
- **Rule of Silence:** Observers must never coach or say "tap that button". The user's hesitation is critical usability data.
- **Pre-Agreed Acceptance Criteria:** Target set beforehand: *"4 of 5 merchants complete fast checkout unaided in under 90 seconds."*
- **Actual STORA Result: 5 of 5 merchants passed, averaging 54 seconds**.

🎙️ PRESENTER SCRIPT (WHAT TO SAY)

> **VERBATIM SPOKEN GUIDE:**
> *"In our UAT, we followed the strict Silent Observation Protocol. We handed the phone to real sari-sari owners, gave them a single goal, and stayed completely silent. We did not rescue them. If they hesitated at checkout, that was our bug to fix. Crucially, we set our pass criteria in advance: 4 out of 5 merchants in under 90 seconds. All 5 completed it unaided in an average of 54 seconds, turning our claim into an objective, verifiable fact."*

SLIDE 12 TOPIC 15: LEGAL & ETHICS RA 10173 Compliance

Testing with people is a privilege — ask first

🎯 SLIDE PURPOSE & CORE MEANING

Applies **Republic Act No. 10173 (Philippine Data Privacy Act of 2012)** to the software testing process. Establishes the Four Pillars of Informed Consent:

- **1. Task Description:** Explicitly state what they will do and the expected duration (10–15 mins).
- **2. Data Collected:** Strictly limited to usability timings and friction logs; zero credit card numbers or financial credentials collected.
- **3. Voluntary Participation:** Participants can stop at any time without penalty or explanation.
- **4. Confidentiality & Right to Erasure:** All participant data is anonymized/de-identified and can be permanently deleted upon request.

🎙 PRESENTER SCRIPT (WHAT TO SAY)

> **VERBATIM SPOKEN GUIDE:**
> *"Testing with real people is not an entitlement—it is a privilege. Under Republic Act 10173, the Philippine Data Privacy Act, you cannot collect user interaction data without prior informed consent. We created a standardized, 1-page consent form in docs/uat/INFORMED_CONSENT_FORM_RA10173.md. Every single participant read and signed this form before touching the app. This demonstrates professional engineering ethics and legal compliance."*

SLIDE 13 TOPIC 16 SECTION HEADER

Accessibility

Topic 16: Accessibility (a11y) & Go-Live Obligations

🎯 SLIDE PURPOSE & CORE MEANING

Establishes accessibility (a11y) not as an optional charity feature, but as a core requirement for real-world software. *"Works for me" is not the same as "works."*

🎙️ PRESENTER SCRIPT (WHAT TO SAY)

> **VERBATIM SPOKEN GUIDE:**
> *"Topic 16 covers Accessibility—often abbreviated as a11y—and Go-Live readiness. In the Philippines, sari-sari store owners are often middle-aged or elderly. They operate outdoors in bright sunlight with glare, or under dim streetlights at night. They might have shaky hands or low vision. If your app only works in an air-conditioned room with 20/20 vision, it will fail on the streets of Iloilo or Manila."*

SLIDE 14 TOPIC 16: THE 5 FIXES

Production Fixes

A few habits cover most of your users

🎯 SLIDE PURPOSE & CORE MEANING

Presents the 5 high-value fixes implemented in STORA based on Web Content Accessibility Guidelines (WCAG 2.1 AA) and their real human impact:

- **1. Contrast (4.5:1 WCAG AA baseline):** STORA achieved **15.8:1 text contrast** so screens are readable under direct noon tropical sun.
- **2. Touch Targets (44×44 dp minimum):** Re-engineered settings and buttons (`_MenuTile` is **54dp tall**) for shaky hands, older adults, or users riding bumpy jeepneys.
- **3. Semantic Labels:** Every `IconButton` has explicit `tooltip` and `semanticLabel` for Android TalkBack screen readers.
- **4. Not Color Alone:** Form errors explicitly print the word "Required" rather than just turning the border red (supports the ~8% of males with color vision deficiency).
- **5. Scalable Text:** Flexible Flutter layouts adapt dynamically when system font scaling is increased up to 200%.

🎙️ PRESENTER SCRIPT (WHAT TO SAY)

> **VERBATIM SPOKEN GUIDE:**
> *"You cannot solve every edge case in one sprint, but these five core habits remove 90% of user barriers. We increased our contrast ratio to 15.8:1, expanded our touch targets to 54dp, added semantic labels to all icon buttons, ensured errors don't rely on red color alone, and validated that layouts survive large text scaling. Accessibility is like curb cuts on sidewalks: built for wheelchairs, but used by everyone with strollers and luggage."*

SLIDE 15 TOPIC 16: TESTING & GAP

Honest Engineering

You can check the basics in fifteen minutes

🎯 SLIDE PURPOSE & CORE MEANING

Shows how easily accessibility can be audited on standard mobile devices without expensive enterprise software. Introduces **The Honest Engineering Gap**.

🎙️ PRESENTER SCRIPT (WHAT TO SAY)

> **VERBATIM SPOKEN GUIDE:**
> *"Testing accessibility does not require expensive labs. We turned on Android TalkBack, closed our eyes, and checked if we could navigate STORA purely through voice cues. We verified our 15.8:1 contrast using standard tools. And most importantly, we bring evidence rather than false claims: We fixed our settings touch targets and contrast, and we openly state our remaining gap: 'Adding audio feedback beeps during camera barcode scanning is our next planned release.' Evaluators value teams who acknowledge their roadmap honestly."*

SLIDE 16 TEAM INTEGRATION

Pipeline Gate

You are the last checkpoint before the world sees it

🎯 SLIDE PURPOSE & CORE MEANING

Summarizes Group 4's strategic leverage across the development lifecycle: building upon Groups 1–3, acting as the storytellers on defense day, and holding the sign-off gate before Group 5 deploys.

🎙️ PRESENTER SCRIPT (WHAT TO SAY)

> **VERBATIM SPOKEN GUIDE:**
> *"Group 4 serves as the final checkpoint. Groups 1 through 3 handed us a working codebase. We diagrammed their components, documented their decisions in Architecture Decision Records, subjected their UI to real sari-sari owners, and enforced accessibility. Group 5 cannot press the deployment button without our sign-off. When we stand before the defense panel, our diagrams and ADRs are what tell the team's engineering story."*

SLIDE 17 DEFENSE EXECUTION

Playbook

The Demo: Explain it, justify it, and prove real people can use it

🎯 SLIDE PURPOSE & CORE MEANING

The Four-Move Demo Playbook for defense day:

- **Move 1 (Diagram):** In 30 seconds, orient the panel with the C4 Container Diagram (Owner App, Customer App, Django, PostgreSQL).
- **Move 2 (ADR):** Read one critical ADR aloud (e.g. ADR-0002: PostgreSQL for utang ACID compliance).
- **Move 3 (UAT):** Present the quantitative result: 5 of 5 store owners in 54s vs 90s criteria, showing the signed RA 10173 form.
- **Move 4 (Accessibility):** Demonstrate 15.8:1 contrast, 54dp targets, TalkBack narration, and state our next milestone.

🎙️ PRESENTER SCRIPT (WHAT TO SAY)

> **VERBATIM SPOKEN GUIDE:**
> *"During defense, we don't wander randomly through screens. We execute four disciplined moves: 30 seconds on the container diagram, reading one ADR aloud, presenting our measured UAT numbers with signed consent, and demonstrating our accessibility fixes. This turns a subjective question into an evidence-based defense."*

SLIDE 18 SCORECARD

Self-Assessment

The checklist you grade yourselves against

🎯 SLIDE PURPOSE & CORE MEANING

Presents the 20-point rubric across all four topics (5 points per topic) that evaluators grade against. Demonstrates that STORA satisfies all 20 criteria.

🎙️ PRESENTER SCRIPT (WHAT TO SAY)

> **VERBATIM SPOKEN GUIDE:**
> *"Here is our self-audit checklist across all 20 criteria. In Topic 13: Context, Container, and ERD diagrams match our active code. In Topic 14: 6 numbered ADRs in docs/adr/ with all 4 required sections. In Topic 15: real merchants, silent observation, recorded timings, and signed consent. In Topic 16: 15.8:1 contrast, 54dp touch targets, TalkBack labels, and font resilience. All 20 boxes are satisfied."*

SLIDE 19 CONCLUSION

Hand-off

A working system, now explained, justified, and proven

🎯 SLIDE PURPOSE & CORE MEANING

Conclusion slide. Re-caps the delivered physical artefacts and formally hands off the deployment pipeline to Group 5 (Day 23: Security, Scalability, Deployment, and Rollback plans).

🎙️ PRESENTER SCRIPT (WHAT TO SAY)

> **VERBATIM SPOKEN GUIDE:**
> *"To conclude: STORA is no longer just working code—it is an explained, justified, and proven platform. We hand over our 6 Architecture Decision Records, C4 blueprints, User Acceptance Testing findings, consent records, and accessibility audit to Group 5. They now own the deployment button to ship STORA safely to the public. Thank you, and we welcome your questions."*